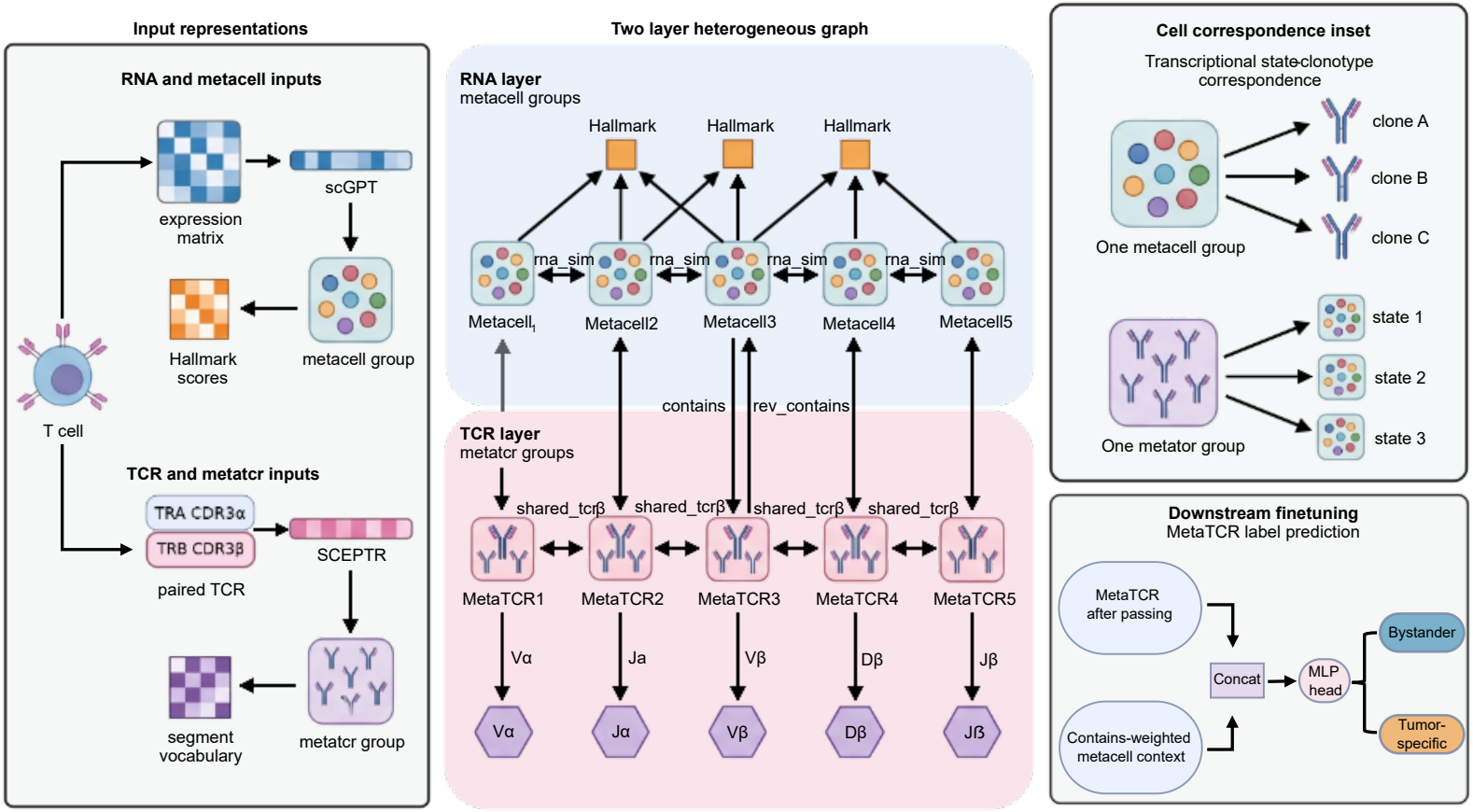

Input representations
RNA and metacell inputs
expression matrix
scGPT
Hallmark scores
metacell group
T cell
TCR and metatcr inputs
TRA CDR3α
TRB CDR3β
paired TCR
SCEPTR
segment vocabulary
metatcr group
Two layer heterogeneous graph
RNA layer
metacell groups
Hallmark
Hallmark
Hallmark
rna_sim
rna_sim
rna_sim
rna_sim
Metacell1
Metacell2
Metacell3
Metacell4
Metacell5
contains
rev_contains
TCR layer
metatcr groups
shared_tcrβ
shared_tcrβ
shared_tcrβ
shared_tcrβ
MetaTCR1
MetaTCR2
MetaTCR3
MetaTCR4
MetaTCR5
Vα
Ja
Vβ
Dβ
Jβ
Vα
Jα
Vβ
Dβ
Jβ
Cell correspondence inset
Transcriptional state-clonotype correspondence
clone A
clone B
clone C
One metacell group
state 1
state 2
state 3
One metator group
Downstream finetuning
MetaTCR label prediction
MetaTCR after passing
Concat
MLP head
Bystander
Tumor-specific
Contains-weighted metacell context